Build Your First Mobile Application Using Apache Cordova

Welcome to the world of mobile application development with the CLI Apache Cordova framework. This article is a tutorial on building a torch application for Android using Web technologies and Apache Cordova.

Mobile apps are everywhere, from smartphones to tablets and now even on smart watches. But developing an app for each specific platform or device can be a tedious task. Imagine a situation where you have an idea for an app and you want to build that app for your target platforms — Android, iOS and Windows. In simple words, it is like developing that same app three times using platform-specific SDKs and APIs. This can be tedious and time consuming. What if there was a way to develop an app that will run on all platforms? This is where Apache Cordova enters the picture.

What is Apache Cordova?

Apache Cordova (earlier known as PhoneGap) is an open source mobile application development framework. It allows programmers to build apps for multiple mobile platforms using standard Web technologies like HTML5, CSS3 and JavaScript instead of relying on platform-specific SDKs and APIs. The resulting apps built using Apache Cordova are hybrid, which means they are neither native mobile apps nor are they Web based. Apache Cordova supports the following platforms - Android, BlackBerry, Firefox OS, iOS, Symbian, Ubuntu Touch, webOS, Windows Phone and Windows 8.

Components of Cordova

A Cordova app has several important components, which are listed below.

WebView: WebView provides the application with its entire user interface, which means you will see your app interface through the WebView component. Think of it as a browser that displays Web pages (user interface). In fact, WebView is an HTML rendering engine, which renders HTML pages.

Web App: This is where your application code is stored. The app is implemented as an HTML Web page and is typically named *index.html*, which points to external files such as CSS, JavaScript and other important resources required for it to run. This component also has a very crucial file called *config.xml*, which stores all-important metadata information about the app, such as its name, description and some specific parameters affecting the working of the app.

Plugins: Plugins are an important component of the Cordova ecosystem. They provide an interface for Cordova and native components to communicate with each other, and are bound to standard device APIs. In other words, they enable you to invoke native code from JavaScript. Apache Cordova maintains such a set of plugins called the Core Plugins, which allow your application to access core device capabilities such as the battery, camera, accelerometer, etc.

Installing Cordova

We are now going to create a mobile app using HTML, CSS, JavaScript and Apache Cordova. Let us first install the Apache Cordova CLI (command-line-interface) tool. Follow the steps shown below:

1. Download and install Node.js (*https://nodejs.org/*). The Cordova command line tool is distributed as an npm package, so we will use Node and npm to install Cordova.
2. Install the Cordova module using the npm utility of Node.js. The Cordova module will be automatically downloaded by the npm utility.
   a. If you are using Windows, run the following command in the command prompt:

   ```
   C:\>npm install -g cordova
   ```

   b. If you are using Linux or OS X, run the following command in the terminal:

   ```
   $ sudo npm install -g cordova
   ```

After the installation, you should be able to run *cordova* on the command line with no arguments, and it should print the help text.